

TECHNICAL PACK

Men's Lightweight Hooded Windbreaker Jacket

Style No. NF-JKT-2026-0417 | Season FW26

Style Number	NF-JKT-2026-0417	PO Number	PO-NF-88231
Brand	Northfield Outdoor	Customer	Atlas Retail Group Ltd.
Season	Fall/Winter 2026 (FW26)	Product Name	Men's Packable Hooded Windbreaker
Version	V03 (Approved for Bulk)	Date	12-May-2026
Factory	Viet Tien Garment JSC — Factory 3	Order Qty	24,500 pcs

CONFIDENTIAL — This document is a simulated technical package generated for software testing and AI-data extraction purposes only. All brand names, codes, and supplier data herein are fictitious and not associated with real companies.

1. PRODUCT OVERVIEW

PRODUCT DESCRIPTION

A lightweight, packable hooded windbreaker designed for transitional fall weather. Features a two-way center front zipper, adjustable drawcord hood, elasticated cuffs, an interior chest stash pocket that doubles as a packing pouch, and a reflective back yoke trim for low-light visibility. Constructed from a 100% recycled nylon ripstop shell with a DWR (durable water repellent) finish. Designed for a regular, slightly relaxed fit suitable for layering.

Garment Type	Outerwear — Hooded Windbreaker Jacket
Gender / Age Group	Men's, Adult
Fit	Regular Fit, Slightly Relaxed
Closure Type	Center Front Two-Way Zipper + Storm Flap Snap
Silhouette	Hip-length, Hooded, Set-in Sleeve
Care Level	Machine Washable, Cold, No Bleach, Line Dry

FABRIC SUMMARY

Component	Fabric	Composition	Weight	Finish
Outer Shell	20D Ripstop Nylon	100% Recycled Polyamide	62 GSM	DWR + Calendared
Lining	Mesh Tricot	100% Polyester	48 GSM	Brushed
Pocket Bag	Taffeta	100% Polyester	55 GSM	Standard

KEY CONSTRUCTION HIGHLIGHTS

Fully taped seams at shoulder & hood for water resistance
Packs into its own chest pocket with carry loop
3M reflective piping on center back yoke

2. TECHNICAL SKETCH (FLAT CAD)

FRONT VIEW
Technical Flat Sketch
(CAD Placeholder)

Fig. 1 — Front View

BACK VIEW
Technical Flat Sketch
(CAD Placeholder)

Fig. 2 — Back View

DETAIL VIEW
Hood / Cuff / Pocket Close-up
(CAD Placeholder)

Fig. 3 — Detail View

CALLOUT REFERENCE MAP

CALLOUT NO.	LOCATION	DESCRIPTION
C1	Center Front	Two-way YKK Vislon zipper, top stop & bottom stop, zipper garage with snap
C2	Hood Opening	Adjustable drawcord with two cord locks, bound edge
C3	Cuffs	Elasticated binding, 25mm width
C4	Hem	Internal drawcord with toggle, two eyelets at side seams
C5	Left Chest	Zippered stash pocket, doubles as packable pouch
C6	Center Back Yoke	Reflective piping trim, 10mm
C7	Center Back Neck	Main label + care label + size label stack

Note: All flat sketches shown are simplified line drawings for sample reference only; refer to released pattern files for exact panel geometry.

3. COLORWAY

COLOR NAME	COLOR CODE	PANTONE TPX	FABRIC CODE	SWATCH	USAGE
Midnight Navy	COL-001	19-4023 TPX	FAB-RPN-20D-NV		Main Colorway (CO1)
Storm Grey	COL-002	17-3914 TPX	FAB-RPN-20D-GY		Secondary Colorway (CO2)
Burnt Orange	COL-003	17-1462 TPX	FAB-RPN-20D-OR		Accent Colorway (CO3)
Forest Green	COL-004	19-5917 TPX	FAB-RPN-20D-GR		Accent Colorway (CO4)
Optic White	COL-005	11-0601 TPX	FAB-RPN-20D-WH		Lining (All COs)

COLORWAY APPLICATION MATRIX

COMPONENT	CO1 NAVY	CO2 GREY	CO3 ORANGE	CO4 GREEN
Shell	COL-001	COL-002	COL-003	COL-004
Lining	COL-005	COL-005	COL-005	COL-005
Zipper Tape	Navy	Grey	Black	Black
Drawcord	Black	Black	Black	Black
Reflective Trim	Silver	Silver	Silver	Silver

Approved colorway for this bulk production run: **CO1 — Midnight Navy** (per PO-NF-88231). All BOM, trim, and packaging data in this document reference CO1 unless noted.

4. BILL OF MATERIALS (BOM)

ITEM	MATERIAL CODE	DESCRIPTION	SUPPLIER	COLOR	UNIT	CONSUMPTION
1	FAB-RPN-20D-NV	20D Ripstop Nylon Shell, DWR	Hengtai Textile Co.	Midnight Navy	m	1.45 m/pc
2	FAB-MTC-48-WH	Mesh Tricot Lining	Hengtai Textile Co.	Optic White	m	1.30 m/pc
3	FAB-TAF-55-BK	Taffeta Pocket Bag Fabric	Hengtai Textile Co.	Black	m	0.20 m/pc
4	ZIP-V5-CF-72	YKK Vislon 2-Way Zipper, 72cm	YKK Vietnam Ltd.	Navy	pc	1 pc/pc
5	ZIP-C3-PKT-18	YKK Coil Zipper, Chest Pocket, 18cm	YKK Vietnam Ltd.	Navy	pc	1 pc/pc
6	TRM-DC-5MM-BK	Drawcord, Round Braided 5mm	Anjie Trims Co.	Black	m	1.10 m/pc
7	TRM-CL-DUAL	Dual-Hole Cord Lock	Anjie Trims Co.	Black	pc	3 pc/pc
8	TRM-EYE-4MM	Metal Eyelet, 4mm	Anjie Trims Co.	Gunmetal	pc	2 pc/pc
9	TRM-SNAP-12L	Snap Button, Size 12L	Anjie Trims Co.	Gunmetal	pc	1 pc/pc
10	TRM-ELS-25MM	Elastic Binding Tape, 25mm	Anjie Trims Co.	Black	m	0.45 m/pc
11	TRM-REF-10MM	3M Reflective Piping, 10mm	3M Authorized Distributor — SafeView	Silver	m	0.55 m/pc
12	LBL-MAIN-WV	Main Label, Woven	Precision Label Co.	Navy/White	pc	1 pc/pc
13	LBL-CARE-PR	Care Label, Printed Satin	Precision Label Co.	White/Black	pc	1 pc/pc
14	LBL-SIZE-WV	Size Label, Woven	Precision Label Co.	White/Black	pc	1 pc/pc
15	THR-TKT80	Polyester Sewing Thread, Tkt 80	Coats Vietnam	Navy	cone	0.012 cone/pc
16	PKG-PBAG-LD	Polybag, LDPE w/ Suffocation Warning	Greenwrap Packaging Ltd.	Clear	pc	1 pc/pc
17	PKG-CTN-RSC	Export Carton, RSC Type	Greenwrap Packaging Ltd.	Brown Kraft	pc	1/25 pc

Consumption figures include standard wastage allowance of 3.5% for fabric and 2% for trims, per Northfield costing standard NF-COST-04.

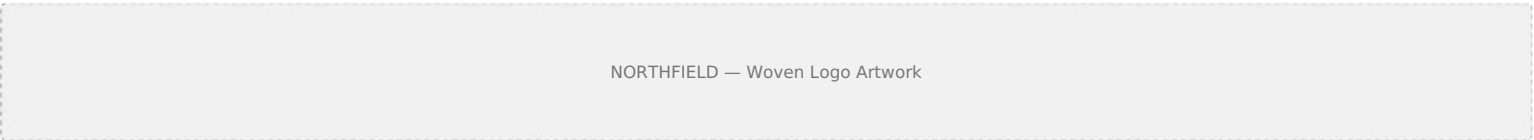
5. TRIM SPECIFICATION

Detailed specification for every trim component used in this style. Reference codes match the BOM and Section 17 Expected Trim List.

TRIM NAME	TRIM CODE	DESCRIPTION	MATERIAL	COLOR	SIZE	SUPPLIER	QTY	UNIT	PLACEMENT
Main Label	TRM-LBL-001	Center-fold woven brand label	Damask Woven Polyester	Navy/White	40 x 25 mm	Precision Label Co.	1	pc	Center Back Neck, below collar seam
Care Label	TRM-LBL-002	Printed satin care & content label, 6-language	Satin Polyester	White/Black	40 x 80 mm	Precision Label Co.	1	pc	Center Back Neck, stacked below Main Label
Size Label	TRM-LBL-003	Woven size indicator (S/M/L/XL/XXL)	Woven Polyester	White/Black	20 x 30 mm	Precision Label Co.	1	pc	Left Side Seam, 100mm below armhole
Hang Tag	TRM-TAG-001	Two-part swing ticket with string	300gsm Recycled Board	Kraft/Navy Print	60 x 100 mm	Precision Label Co.	1	pc	Attached to Main Zipper Pull via plastic snap
Price Ticket	TRM-TAG-002	Adhesive price/SKU ticket	Coated Paper	White	40 x 60 mm	Precision Label Co.	1	pc	Affixed to Hang Tag reverse side
Drawcord	TRM-DC-5MM-BK	Round braided drawcord, tipped ends	Polyester Braid	Black	5mm dia.	Anjie Trims Co.	1.10	m	Hood opening & hem channel
Eyelet	TRM-EYE-4MM	Metal eyelet with washer	Brass, Gunmetal Plated	Gunmetal	4mm I.D.	Anjie Trims Co.	2	pc	Hem, side seams (drawcord exit points)
Zipper (Main)	ZIP-V5-CF-72	2-way Vislon zipper, auto-lock slider	POM Tooth, Polyester Tape	Navy	72 cm	YKK Vietnam Ltd.	1	pc	Center Front Opening
Zipper (Pocket)	ZIP-C3-PKT-18	Coil zipper, reverse-coil, water-resistant	Nylon Coil, Polyester Tape	Navy	18 cm	YKK Vietnam Ltd.	1	pc	Left Chest Pocket
Slider (Pull)	TRM-SLD-001	Rubberized zipper pull extension	TPU + Metal Loop	Black	45mm length	YKK Vietnam Ltd.	2	pc	Main Zipper + Pocket Zipper sliders
Button (Storm Flap)	TRM-BTN-001	Flat 4-hole sew-through button	Polyester Resin	Navy	15L	Anjie Trims Co.	0	pc	N/A — replaced by snap on this style
Snap (Storm Flap)	TRM-SNAP-12L	Prong snap fastener, 4-part	Brass, Gunmetal Plated	Gunmetal	12L	Anjie Trims Co.	1	pc	Storm flap above main zipper top stop
Elastic Binding	TRM-ELS-25MM	Knit elastic binding tape	Polyester/Spandex	Black	25mm width	Anjie Trims Co.	0.45	m	Cuff openings (both sleeves)
Velcro (Cuff Tab)	TRM-VEL-20MM	Hook & loop adjustable tab	Nylon	Black	20mm width	Anjie Trims Co.	0	pc	N/A — not used this style; elastic cuff only
Polybag	PKG-PBAG-LD	LDPE bag with suffocation warning print	LDPE, 0.04mm	Clear	350 x 450 mm	Greenwrap Packaging Ltd.	1	pc	Final folded garment packaging
Carton Sticker	PKG-STK-001	Self-adhesive carton ID label	Coated Paper	White/Black	100 x 150 mm	Greenwrap Packaging Ltd.	2	pc	Two opposite side panels of export carton

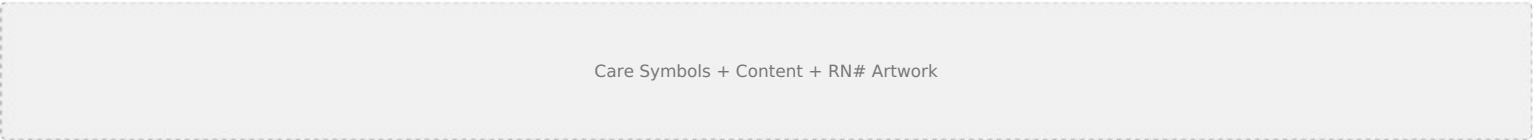
6. LABEL SPECIFICATION

MAIN LABEL



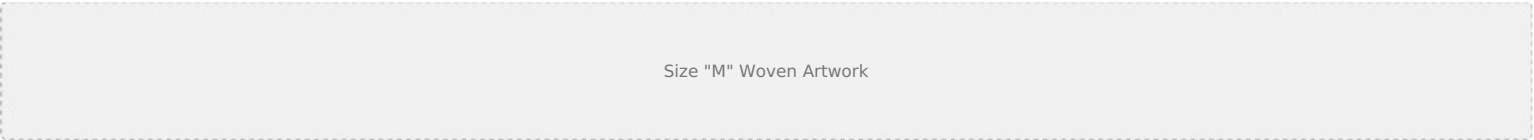
Type Center-fold Woven, Damask
Size 40 x 25 mm
Attachment Sewn into back neck seam

CARE LABEL



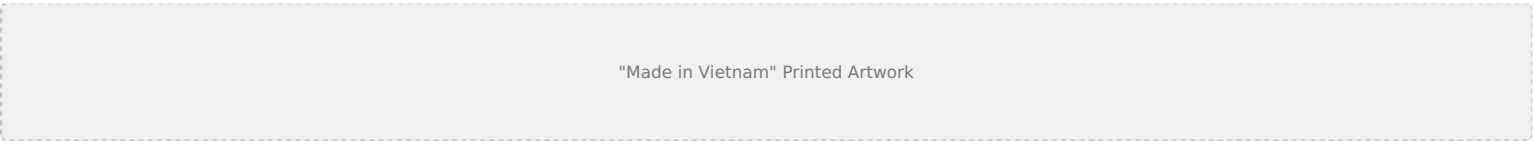
Type Printed Satin, Folded
Size 40 x 80 mm
Languages EN / FR / ES / DE / PT / ZH

SIZE LABEL



Type Woven, Single Layer
Size 20 x 30 mm
Position Left side seam, below Country of Origin tag

COUNTRY OF ORIGIN LABEL



Type Printed Satin
Size 20 x 40 mm
Required Text MADE IN VIETNAM

LABEL STACK ORDER (CENTER BACK NECK)

POSITION (TOP TO BOTTOM)	LABEL	NOTES
1	Main Label	Sewn into neck binding seam
2	Care Label	Folded, 4-sided print, sewn below main label
3	Country of Origin	Sewn at side seam, NOT neck stack — see size label note

7. FABRIC SPECIFICATION

Component	Composition	GSM	Construction	Finish	Supplier
Shell Fabric	100% Recycled Polyamide (Nylon)	62 GSM	20D Ripstop, Plain Weave	DWR (C0 PFC-Free) + Calendared	Hengtai Textile Co.
Lining Fabric	100% Polyester	48 GSM	Warp Knit Tricot Mesh	Brushed, Anti-static	Hengtai Textile Co.
Pocket Bag Fabric	100% Polyester	55 GSM	Taffeta, Plain Weave	Standard, Soft Hand	Hengtai Textile Co.

PERFORMANCE TEST REQUIREMENTS

Test	Standard	Requirement	Result (V03 Sample)
Water Repellency (Spray Test)	AATCC 22	Rating ≥ 90	Pass — 90
Hydrostatic Pressure	ISO 811	≥ 800 mm	Pass — 1100 mm
Tear Strength (Shell)	ASTM D1424	≥ 8 N	Pass — 11.2 N
Colorfastness to Rubbing	ISO 105-X12	Grade ≥ 4	Pass — Grade 4-5
Colorfastness to Washing	ISO 105-C06	Grade ≥ 4	Pass — Grade 4
Pilling Resistance	ISO 12945-2	Grade ≥ 3-4	Pass — Grade 4

8. MEASUREMENT SPECIFICATION

SIZE CHART (IN CM, BODY MEASUREMENT)

CODE	POINT OF MEASURE	TOLERANCE	S	M	L	XL	XXL
A	Chest Width (1" below armhole)	± 1.0	54.0	57.0	60.0	63.0	66.0
B	Body Length (HPS to Hem)	± 1.0	69.0	71.0	73.0	75.0	77.0
C	Sleeve Length (CB to Cuff)	± 0.8	84.5	86.5	88.5	90.5	92.5
D	Shoulder Width	± 0.6	46.0	48.0	50.0	52.0	54.0
E	Hem Width	± 1.0	50.0	53.0	56.0	59.0	62.0
F	Cuff Width (Relaxed)	± 0.5	9.0	9.5	10.0	10.5	11.0
G	Hood Height	± 0.6	34.0	34.5	35.0	35.5	36.0
H	Front Zipper Length	± 0.5	68.0	70.0	72.0	74.0	76.0

Measurements taken with garment laid flat, unstretched, per Northfield Measurement Standard NF-MS-02. All tolerances apply at finished garment, post-wash where applicable.

GRADING NOTES

Grade rule increment: 3.0 cm per size at chest; asymmetric grading applied at hood height (+0.5cm per size) to maintain proportion. Refer to digital grading file NF-GRD-0417-V03.xlsx for full point-by-point grade rules.

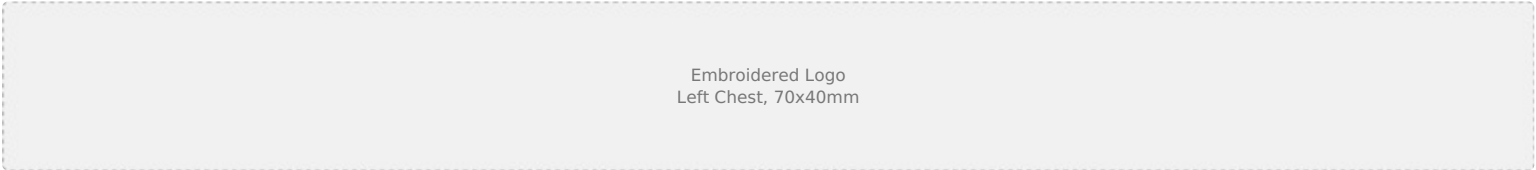
9. CONSTRUCTION DETAILS

AREA	SEAM TYPE	STITCH TYPE	SPI	THREAD
Side Seams	Flat-Felled (SSa-2)	301 Lockstitch	10-12	Tkt 80 Polyester
Shoulder Seams	Taped Seam, Bound	401 Chainstitch + Tape	9-11	Tkt 80 Polyester
Hood Seam	Taped Seam, Bound	401 Chainstitch + Tape	9-11	Tkt 80 Polyester
Hem & Cuff Binding	Bound Seam (BSc-1)	406 Coverstitch	10-12	Tkt 80 Polyester
Pocket Attachment	Lapped Seam (LSa-1)	301 Lockstitch	10-12	Tkt 80 Polyester
Topstitching (Visible)	Edge Stitch, 3mm	301 Lockstitch	8-10	Tkt 60 Polyester (Contrast Navy)

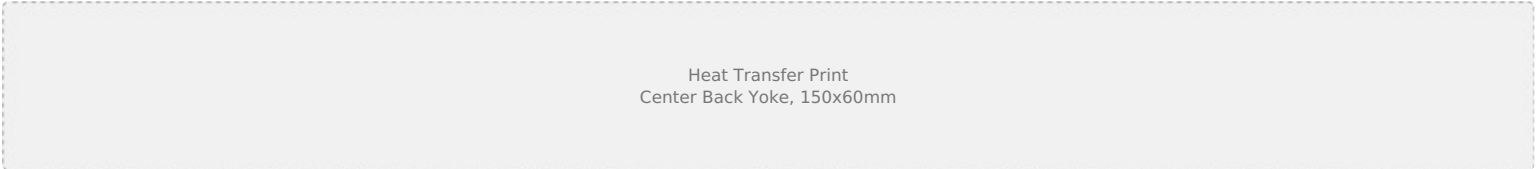
SEAM SEALING REQUIREMENT

All shoulder, hood, and yoke seams must be heat-taped with 20mm waterproof seam tape per QA checklist item 4.2. Tape adhesion to be verified by 30-minute rain simulation test before bulk approval.

10. ARTWORK & PRINTING



Logo — Embroidery



Reflective Print Placement



Hardware Branding

ELEMENT	TECHNIQUE	PLACEMENT	SIZE	COLORS
Brand Logo	Flat Embroidery, 8 Stitch Density	Left Chest, 8cm below shoulder seam	70 x 40 mm	White, Silver Grey
Reflective Graphic	Heat Transfer Film	Center Back Yoke	150 x 60 mm	Silver Reflective
Zipper Pull Branding	Debossed Logo	Main & Pocket Zipper Pulls	10 x 10 mm	Tonal

11. PACKAGING

Polybag	LDPE, 0.04mm, clear, suffocation warning printed (per CPSC/EN71), self-adhesive flap seal, size 350 x 450mm
Folding Method	Standard 3-fold flat pack with tissue insert, hang tag visible through bag
Carton	RSC export carton, 5-ply corrugated, 60 x 40 x 35 cm, 25 pcs/carton, gross weight ≈ 11.8kg
Carton Sticker	Affixed to two opposite long panels; includes Style No., PO No., Color, Size Breakdown, Carton No., Total Cartons
Carton Marking	Side marks: Customer Name, Destination Port, PO Number, Carton No./Total, "Made in Vietnam"



12. BARCODE / SKU

SIZE	SKU	UPC	EAN-13	QR CODE
S	NF-JKT-0417-NV-S	8 12345 67901 2	4 901234 567890	QR
M	NF-JKT-0417-NV-M	8 12345 67902 9	4 901234 567906	
L	NF-JKT-0417-NV-L	8 12345 67903 6	4 901234 567913	
XL	NF-JKT-0417-NV-XL	8 12345 67904 3	4 901234 567920	
XXL	NF-JKT-0417-NV-XXL	8 12345 67905 0	4 901234 567937	

QR code links to digital product passport: care instructions, material traceability, and recycling guidance, per EU Digital Product Passport pilot requirement.

13. SUPPLIER INFORMATION

SUPPLIER	COMPONENT SUPPLIED	LOCATION	CONTACT
Hengtai Textile Co.	Shell, Lining, Pocket Fabric	Hangzhou, China	sales@hengtai-textile.example.com
YKK Vietnam Ltd.	Zippers & Sliders	Bien Hoa, Vietnam	orders@ykk-vn.example.com
Anjie Trims Co.	Drawcord, Eyelets, Snaps, Elastic	Dongguan, China	info@anjietrims.example.com
Precision Label Co.	Labels, Hang Tags, Tickets	Ho Chi Minh City, Vietnam	contact@precisionlabel.example.com
SafeView (3M Distributor)	Reflective Piping	Shenzhen, China	sales@safeview.example.com
Greenwrap Packaging Ltd.	Polybag, Cartons, Stickers	Hai Phong, Vietnam	orders@greenwrap.example.com
Coats Vietnam	Sewing Thread	Ho Chi Minh City, Vietnam	vn.sales@coats.example.com
Viet Tien Garment JSC	Cut, Make, Trim (CMT Factory)	Nam Dinh, Vietnam	merchandising@viettien.example.com

14. QA CHECKLIST

#	CHECKPOINT	REQUIREMENT	STATUS
1	Stitch Density / SPI	Per construction spec, Section 9	PASS
2	Seam Tape Adhesion	No peel after 30-min rain simulation	PASS
3	Label Placement Accuracy	± 5mm of specified position	PASS
4	Zipper Function	Smooth 2-way operation, no snag, 20-cycle pull test	PASS
5	Colorfastness	Per Section 7 test requirements	PASS
6	Measurement Compliance	Within tolerance per Section 8	PASS
7	Hardware Safety (Snap/Eyelet)	Pull test ≥ 90N, no sharp edges	PASS
8	Packaging Compliance	Suffocation warning present, correct carton marks	PASS

15. REVISION HISTORY

VERSION	DATE	CHANGE DESCRIPTION	UPDATED BY
V01	02-Mar-2026	Initial tech pack release for proto sample	A. Lindqvist (Design)
V02	09-Apr-2026	Updated cuff from velcro tab to elastic binding; revised BOM	T. Nguyen (Tech Design)
V03	12-May-2026	Approved for bulk; finalized colorway CO1, added reflective trim spec	T. Nguyen (Tech Design)

16. EXPECTED TRIM LIST

This table consolidates all trim components specified throughout this document (Sections 4 & 5). It is provided as the reference output that an AI Agent should reproduce when generating a Trim List from this Tech Pack.

NO	TRIM	DESCRIPTION	COLOR	SIZE	SUPPLIER	QTY	UNIT
1	Main Label	Center-fold woven brand label	Navy/White	40 x 25 mm	Precision Label Co.	1	pc
2	Care Label	Printed satin care & content label, 6-language	White/Black	40 x 80 mm	Precision Label Co.	1	pc
3	Size Label	Woven size indicator	White/Black	20 x 30 mm	Precision Label Co.	1	pc
4	Hang Tag	Two-part swing ticket with string	Kraft/Navy Print	60 x 100 mm	Precision Label Co.	1	pc
5	Price Ticket	Adhesive price/SKU ticket	White	40 x 60 mm	Precision Label Co.	1	pc
6	Drawcord	Round braided drawcord, tipped ends	Black	5mm dia.	Anjie Trims Co.	1.10	m
7	Cord Lock	Dual-hole cord lock	Black	—	Anjie Trims Co.	3	pc
8	Eyelet	Metal eyelet with washer	Gunmetal	4mm I.D.	Anjie Trims Co.	2	pc
9	Zipper (Main)	2-way Vislon zipper, auto-lock slider	Navy	72 cm	YKK Vietnam Ltd.	1	pc
10	Zipper (Pocket)	Coil zipper, reverse-coil, water-resistant	Navy	18 cm	YKK Vietnam Ltd.	1	pc
11	Slider (Pull)	Rubberized zipper pull extension	Black	45mm length	YKK Vietnam Ltd.	2	pc
12	Snap (Storm Flap)	Prong snap fastener, 4-part	Gunmetal	12L	Anjie Trims Co.	1	pc
13	Elastic Binding	Knit elastic binding tape	Black	25mm width	Anjie Trims Co.	0.45	m
14	Reflective Piping	3M reflective piping trim	Silver	10mm width	SafeView (3M Distributor)	0.55	m
15	Polybag	LDPE bag with suffocation warning print	Clear	350 x 450 mm	Greenwrap Packaging Ltd.	1	pc
16	Carton Sticker	Self-adhesive carton ID label	White/Black	100 x 150 mm	Greenwrap Packaging Ltd.	2	pc

Items intentionally not used on this style (Button, Velcro) are documented in Section 5 with Qty = 0 and should be excluded from the final Trim List output above.